So, to start, I worked on the first enhancement of my inventory management app focusing on software design and engineering. This app was originally created for my CS 360 Mobile Architecture and Design class.

This was included in my portfolio because it needed work and I enjoy app development. Career-wise, I want to pursue app development, so this project felt easy to incorporate into my actual e-portfolio. I want this project to highlight my skill in design and code architecture. I follow coding standards, practice safe coding practices, and show enhancement in ability from prior classes. In this enhancement, I mainly changed the codebase for the core functionality from Java to Kotlin, introduced a repository that will be used in enhancement two and three, introduced ViewModel, and moved logic away from ActivityMain to the new files created. Overall, the app is more modernized, which was a primary goal for this enhancement, and is now being set up for the next enhancements.

Yes, I would say I met the course outcomes I planned to meet I made in Module One. It demonstrates that I use well-founded techniques, skills, and tools in computing practices to implement solutions that deliver value. The change from Java to Kotlin, the addition of a repository, and the creation of the ViewModel demonstrate this because they are meant to improve organization while keeping the same functionality of the existing program, while also setting up further enhancements. The code reviews and feedback from my instructor have also helped me stay on track and focus on the key areas I want to enhance.

As for updated to my plans, it is more focused and less general. I originally wanted to add several major features in this enhancement, however, my instructor and I both decided it would be best to separate them into more achievable goals. My plan moving forward is to focus on

adding quantity-based sorting and adding a lot-stock filter that the user can choose, aiming for a speed of O(n log n) and O(n), respectively. Enhancement three will focus on replacing the SQLite database with Room, which is what I am actively setting the project up for now.

While working on this project, I would often look back to see how different I approach the challenges I face now. As I create new files and enhance old ones, I see the project as a whole instead of just individual parts now. I usually need to reflect often to stay on track for the entire project, but this time, it just clicked. Maybe it is due to working on this project before? I am unsure. A problem I have been facing is using both Kotlin and Java in this project. It is the first time I have used multiple code bases where they work together. During development, when I would automatically convert the java files to Kotlin, I would need to go in and change a lot of small additives in order to get them to work. For example, there were countless numbers of "!!" in the file, as well as most of my variables being set to null. The class names were also different. I would say this is the biggest realization I had, that changing languages does not automatically improve the design or structure of the project. For example, I was able to shave off about 50 lines of code in the big files and about 20 lines of code in the other files. This is nice, but it came at the cost of needing to go in and change each function or line of code to ensure it works.